

SHAGUFTA ZAINAB

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Education

S.A. Engineering College, Chennai

2022 - 2026

Bachelor of Engineering in Electronics & Communication

CGPA: 8.8

Smt. Chandabai Pagariya Jain Mat. Hr. Sec. School

Grad. 2022

Science Stream (Computer Science with Maths)

85.00%

Skills

Networking: TCP/IP, IP addressing, subnetting, routing (static, dynamic, default) , switching (L2/L3), VLANs, network troubleshooting, hands-on practice using Cisco Packet Tracer (CCNA Training)

Programming: Working knowledge of Java & Python

Tools: Cisco Packet Tracer, Arduino IDE, Xilinx Vivado, Microsoft Office

Interpersonal: Analytical problem-solving, Clear technical communication, Time Management, Leadership, Team Management, Critical Thinking

Languages: English, Hindi, Tamil, Urdu

Experience

Simpson & Co Pvt Ltd, Intern ([Certificate](#))

June 2025 – August 2025

- Worked with SAP to manage and maintain enterprise digital data, ensuring accuracy and consistency across records.
- Identified discrepancies and resolved data issues through structured analysis and documentation.
- Collaborated across teams to improve reporting accuracy and workflow efficiency, strengthening problem-solving and cross-functional communication.

Projects

Network Simulation & Routing Labs (CCNA Practice)

2026

- Built and tested multiple network topologies using Cisco Packet Tracer.
- Configured VLANs, routing protocols, IP addressing, and switching scenarios.
- Performed connectivity validation and troubleshooting across simulated enterprise networks.

Design of a High-Performance ALU for 32-bit RISC-V Processor

2025 - 2026

- Designed a 32-bit ALU for a single-cycle RISC-V processor in Verilog HDL using Xilinx Vivado, replacing RCA and array multiplier with Brent-Kung adder and Vedic multiplier.
- Reduced ALU delay to 9.89 ns, power to 0.598 W, and area to 106 LUTs using parallel prefix carry computation and parallel partial-product generation.

Smart Cradle System

2025

- Engineered an automated smart cradle using NodeMCU (ESP8266), PIR sensor, servo motor, and DHT11, enabling automatic swing, temperature monitoring, and buzzer-based safety alerts.

Certifications

- Cisco Networking Academy — **CCNA** (Modules 1–3) — In Progress
- Python—Dynamic CAD
- Enhancing Soft Skills and Personality—NPTEL (Top 5 % scorer)
- IoT and IIoT—Techknots Academy LLP

Achievements & Activities

Department Topper

- Awarded Department Topper twice during the B.E. program by maintaining exceptional academic performance and ranking highest in the department.

Intercollegiate Hackathon Participation – HackAIThon 2025

- Contributed by ideating the solution approach and debugging code using AI tools during the HackAIThon intercollegiate competition at Stella Maris College, collaborating in a fast-paced problem-solving environment.